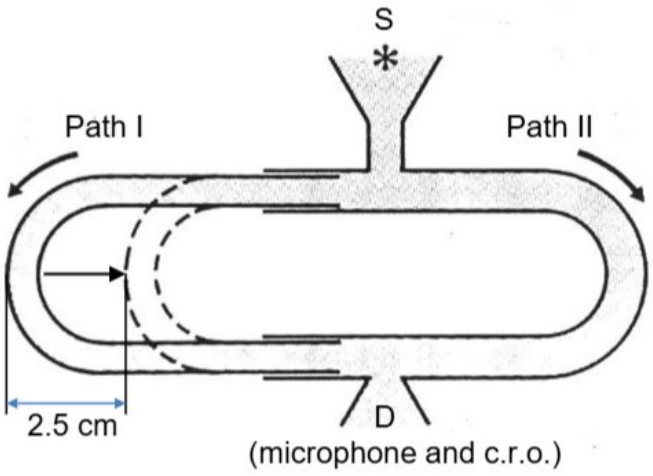
16 A sound source S emits a continuous sound wave of constant frequency into a tube. The sound wave travels along two paths, I and II, before recombining at D where a microphone connected to a cathode ray oscilloscope (c.r.o) is placed.



The sliding section in Path I is adjusted until a minimum sound intensity is detected at D. The sliding section is then slowly pushed inwards. The next maximum sound intensity is detected when the sliding section has moved through a horizontal distance of 2.5 cm.

Given that the speed of sound in the tube is 340 m s^{-1}, what is the frequency of the sound wave?

A 1.7 kHz **B** 3.4 kHz **C** 6.8 kHz **D** 14 kHz